

Name: Priyanshu Digari
Roll no. : 89

Assignment - 04
Image Processing and Pattern
Recognition
TCS-361

Q1.) In a factory that produces bolts, three machines—A, B, and C—are responsible for manufacturing 25%, 35%, and 40% of the total bolts, respectively. The defect rates of these machines differ: machine A produces 5% defective bolts, machine B produces 4% defective bolts, and machine C produces 2% defective bolts. A bolt is randomly selected from the overall production, and it is found to be defective. What is the probability/likelihood that this

defective bolt was produced by machine B?

Ans.) $P(A)=0.25$, $P(B)=0.35$, $P(C)=0.40$
 $P(D/A)=0.05$, $P(D/B)=0.04$,
 $P(D/C)=0.02$

$$P(B/D) = [P(D/B)P(B)] / [P(D/A)P(A) + P(D/B)P(B) + P(D/C)P(C)]$$

$$\begin{aligned} &= (0.04 \times 0.35) / [(0.05 \times 0.25) + \\ & (0.04 \times 0.35) + (0.02 \times 0.40)] \\ &= 0.014 / (0.0125 + 0.014 + 0.008) \\ &= 0.014 / 0.0345 \\ &= 0.4058 \end{aligned}$$

Q82.) Two cards are drawn from a pack of 52. Find the probability that one may be queen and other a king.

Ans.) Total ways = $52C2 = 1326$
Favorable = $(4 \text{ queens} \times 4 \text{ kings}) = 16$

$$\text{Probability} = 16 / 1326 = 8 /$$

Q83.) Derive Binomial Distribution and give its characteristics. Explain K-means algorithm.

Ans.) Binomial Distribution:

$$P(X = k) = nCk p^{**k} (1-p)^{**}(n-k)$$

Characteristics:

Mean = np

Variance = $np(1-p)$

Fixed number of trials (n)

Two outcomes: success/failure

Independent trials

K-means Algorithm:

1. Choose k clusters
2. Initialize centroids
3. Assign points to nearest centroid
4. Update centroids (mean of cluster)